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Relationships between undegraded and physically effective fiber in diets for lactating Holstein dairy cows

W. A. Smith,[1] K. Ishida,[2] J. W. Darrah,[1] H. M. Dann,[1] C. S. Ballard,[1] M. D. Miller,[1] S. Y. Morrison,[1]* and R. J. Grant[1]
[1]William H. Miner Agricultural Research Institute, Chazy, NY 12921
[2]Zen-Noh National Federation of Agricultural Cooperative Associations, Tokyo, Japan 100-6832

ABSTRACT

We measured the effect of physically effective NDF (peNDF) in diets containing lower or higher concentrations of undegraded NDF at 240 h of in vitro fermentation (uNDF240) on DMI, milk production and composition, chewing behavior, ruminal responses, and nutrient digestibility. Sixteen Holstein cows (8 ruminally cannulated) averaging 123 ± 9 (SD) DIM were used in a replicated 4 × 4 Latin square design with 4-wk periods. Cows were fed TMR that differed in uNDF240 and peNDF content primarily by changing forage-to-concentrate ratio and particle size of timothy hay. Treatments were arranged as a 2 × 2 factorial arrangement consisting of (1) lower uNDF240 (LU; 8.9% of TMR DM) or higher uNDF240 (HU; 11.4% of TMR DM) and (2) lower peNDF (LP; 18.6%–20.1% of TMR DM) or higher peNDF (HP; 21.8%–22.0% of TMR DM). Physically effective uNDF240 (peuNDF240) was calculated as the product of the dietary physical effectiveness factor (pef; % of TMR DM retained on a ≥1.18-mm sieve with vertical dry sieving) and uNDF240 (as a percentage of DM) with the purpose of combining particle size and undegradability of fiber into one measure. Dietary peuNDF240 concentrations were 5.4% (LU-LP), 5.8% (LU-HP), 5.9% (HU-LP), and 7.1% (HU-HP) of TMR DM. Cows were housed in individual tiestalls, fed TMR once daily, and milked 3×/d. Data were analyzed as a factorial arrangement of treatments within a replicated Latin square design using the MIXED procedure of SAS (ver. 9.4; SAS Institute Inc., Cary, NC). The model included the fixed effects of uNDF240, peNDF, uNDF240 × peNDF, period within replicated square, and square. Cow within square was a random effect. There was an interaction between dietary peNDF and uNDF240, with cows fed the HU-HP diet consuming 10% less DM (24.9 kg/d) than cows fed the HU-LP (27.4 kg/d) or either LU diet (27.4 kg/d). Diets higher in peNDF content elicited greater peNDF intake. However, there was an interaction between uNDF240 and peNDF for uNDF240 intake driven by the HU-LP diet (3.1 kg/d) compared with the HU-HP (2.9 kg/d) and LU diets (2.4 kg/d). Intake of peuNDF240 reflected dietary pef and uNDF240 content, as expected. There was an effect of uNDF240 and peNDF on ECM production, and it tracked with dietary peuNDF240 (47.5, 45.2, 46.5, and 44.1 kg/d for diets LU-LP, LU-HP, HU-LP, and HU-HP, respectively). Mean ruminal pH increased, total VFA concentration (millimolar) decreased, and acetate-to-propionate ratio increased with higher dietary uNDF240, but peNDF content had no effect. Rumination time (min/d) was unaffected by diet, although eating time (min/d) was increased by both uNDF240 and peNDF content. An interaction between uNDF240 and peNDF resulted in lower dietary peNDF reducing eating and rumination time (min/kg of DMI) primarily for the HU diets. The HU diets increased the ruminal pool size of NDF and uNDF240 versus the LU diets, slowed NDF ruminal turnover rate, and increased apparent total-tract digestibility of potentially degradable NDF (NDF − uNDF240). Responses in DMI, ECM, chewing behavior, and ruminal characteristics reflected dietary particle size and fiber undegradability as measured by dietary uNDF240, peNDF, and peuNDF240 content.
Key words: particle size, physically effective fiber, undegraded fiber, eating, ruminating

INTRODUCTION

Chemical measurement of NDF does not account for the physical or degradability characteristics of fiber (Van Soest, 1994). Mertens (1997) incorporated particle size of fiber into the NDF system with the development of physically effective NDF (**peNDF**). Development of undegraded NDF (**uNDF**) after 240 h of in vitro fermentation (OM basis; **uNDF240om**) allowed for laboratory measurement of indigestible NDF for individual feed-

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*Corresponding author: morrison@whminer.com

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stuffs and TMR (Raffrenato et al., 2018). Because both factors affect DMI and lactational performance, a better understanding of the relationship between dietary uNDF240om and peNDF would facilitate ration formulation to satisfy the fiber requirements of lactating dairy cows. Little research has been conducted evaluating the interaction between uNDF240om and peNDF and the effect on ruminal digesta turnover, chewing behavior, and DMI.

Physically effective NDF is the fraction of dietary fiber with sufficient particle size to stimulate chewing (ruminating and eating) and create a well-formed ruminal digesta mat (Mertens, 1997). Physically effective NDF is commonly measured by multiplying the NDF content of a feed by the fraction of particles retained on or above the 1.18-mm sieve for dried samples (measured with vertical sieving; Mertens, 1997). This fraction of larger particles is termed the physical effectiveness factor (**pef**; Mertens, 1997). Greater forage and dietary peNDF have been associated with more chewing activity, lower DMI, and higher ruminal pH in lactating cows (Mertens, 1997; Beauchemin, 2018).

Using undegraded NDF at 240 h of in vitro fermentation (**uNDF240**), the potentially degradable NDF (**pdNDF**) fraction of NDF can be calculated (i.e., NDF − uNDF240) and ultimately the fractional rate of degradation for pdNDF may be determined (Waldo et al., 1972). Aside from its obvious use in determining pdNDF, uNDF240om influences ruminal fill, ruminal fiber degradation and passage dynamics, and the physical effectiveness of forages (Cotanch et al., 2014; Harper and McNeill, 2015; Miller et al., 2021). Diets with greater uNDF240om content elicit a reduction in DMI associated with slower ruminal turnover of NDF and consequently lower output of milk and milk components (Miller et al., 2021). With corn silage–based diets, Miller et al. (2021) concluded that dietary uNDF240om content greater than about 10% of ration DM constrained DMI due to gut fill. Likewise, for corn silage–based diets ranging from 9.0% to 11.1% of ration DM, Vieira et al. (2025) observed that each 1 percentage unit increase in uNDF (measured in situ at 288 h of fermentation) was associated with 0.59 kg/d reduction in DMI and 0.75 kg/d reduction in FCM yield.

Combining a measure of particle size such as pef with uNDF240om may allow for better prediction of DMI. We have termed this combination physically effective uNDF240 (**peuNDF240**; Smith et al., 2018). The few studies conducted thus far indicate that peuNDF240 could be useful as a metric for monitoring the effect of reducing particle size of forages low in NDF degradability to maintain milk yield and efficiency of milk production (Serva et al., 2021). Serva et al. (2021) found the median dietary uNDF240om to be 8.3% of ration DM across 22 dairy farms feeding corn silage, grass silage, and alfalfa hay as their primary forages. Similarly, Kerwin et al. (2022) found the mean uNDF240 in high cow rations to be 9.7% with a range of 6.1% to 15.6% of ration DM for 72 dairy herds in the northeastern United States. To date, no study has evaluated the interaction of particle size with uNDF240 within the range of uNDF240 between the mean and higher dietary content where DMI may most likely be constrained and where many diets fall given the common challenges of harvesting forages at the correct stage of maturity. Of the studies that have reported peuNDF240, Serva et al. (2021) assessed alfalfa and corn silage whereas Miller et al. (2021) and Vieira et al. (2025) assessed primarily corn silage indicating the potential usefulness of this metric across forage types.

The objective of this study was to evaluate the effects of feeding higher and lower dietary concentrations of uNDF240om and peNDF on DMI, lactational performance, chewing behavior, and the ruminal environment of lactating Holstein dairy cows. We hypothesized an interaction between uNDF240om and peNDF whereby smaller particle size would elicit greater DMI and ECM responses from cows fed higher uNDF240 diets with smaller effects at lower, more moderate dietary uNDF240om. We further hypothesized that higher dietary uNDF240om and peNDF concentrations would result in greater ruminal pH, more chewing per kilogram of DMI, slower rumen NDF turnover, and less DMI compared with lower uNDF240om and peNDF diets. Finally, we hypothesized that dietary peuNDF240 (i.e., pef × uNDF240) would track meaningfully with DMI, ECM, chewing behavior, and ruminal responses.

MATERIALS AND METHODS

Experimental Design, Treatments, and Management of Cows

The study was conducted at the William H. Miner Agricultural Research Institute (Chazy, NY) in the Charles J. Sniffen Dairy Research and Education Complex. All experimental procedures involving animals were approved by the William H. Miner Agricultural Research Institute Animal Care and Use Committee. Sixteen multiparous Holstein cows (8 ruminally cannulated) were used in a replicated 4 × 4 Latin square design study with 28-d periods (squares were conducted concurrently) and a factorial arrangement of treatments. The first 19 d served as adaptation and the last 9 d served as the collection period. This design (i.e., animal numbers and expected intake variation) and length of experimental period have been shown previously to allow for measurement of animal response to dietary NDF degradability (Oba and Allen, 2000; Miller et al., 2021). Change-over designs such as Latin squares are as accurate as continuous designs in measuring DMI and milk production responses, except for extreme treatments that elicit extensive mobilization of body fat and protein (Huhtanen and Hetta, 2012). At the start of the study, animals were blocked by fistulation status, DIM (mean ± SD; 123 ± 9), milk production (53.0 ± 4.5 kg/d), and parity (2.4 ± 0.7).

Four diets in which the primary forage was corn silage were formulated to contain either a lower or higher concentration of uNDF240om and a lower or higher concentration of peNDF. The different amounts of uNDF240om were achieved primarily by varying the forage-to-concentrate ratio of the diet (Table 1). A hammer mill forage processor (Haybuster; DuraTech Industries International Inc., Jamestown, ND) with 2 different sieve sets (7.62 and 5.08 cm; 1.27 and 0.95 cm) was used to achieve 2 particle sizes of dry timothy hay. In addition, for the lower forage diets, a portion of timothy hay was replaced with pelleted beet pulp to balance the fiber fractions across the 4 diets. The 4 dietary treatments were (1) lower uNDF240om and lower peNDF concentrations (**LU-LP**), (2) lower uNDF240om and higher peNDF concentrations (**LU-HP**), (3) higher uNDF240om and lower peNDF concentrations (**HU-LP**), and (4) higher uNDF240om and higher peNDF concentrations (**HU-HP**). Diets were formulated using a commercial ration formulation platform (AMTS. Cattle.Professional, Agricultural Modeling & Training systems LLC, Groton, NY; version 4.8) with the Cornell Net Carbohydrate and Protein System (**CNCPS**) biology (v 6.5.5, Cornell University, Ithaca, NY). Inputs used for dietary formulation included 28.1 kg/d DMI, 52.2 kg/d milk with 3.80% fat and 3.05% true protein, and 726 kg of BW. The lower uNDF240om diets contained 46.8% forage and the higher uNDF240om diets contained 60.5% forage on a DM basis (Table 1).

Cows were fed treatment TMR for ad libitum intake (approximately 1.05 × expected intake) at 1400 h once daily (Calan Data Ranger; American Calan Inc., Northwood, NH). Cows were housed in tiestalls equipped with individual feed boxes and water bowls. Cows were milked 3 times daily (0430, 1230, and 2030 h) in a double-12 parallel milking parlor (Xpressway Parallel Stall System; Bou-Matic, Madison, WI). Using Hobo data loggers (Onset, Bourne, MA), temperature and relative humidity were recorded at 15-min intervals throughout the study.

Data Collection, Sampling Procedures, and Analytical Methods

DMI, Milk Yield and Composition, Efficiency, BW, and BCS. Dry matter intake was determined by recording feed offered and refused on d 20 to 28 for each cow during each period. Samples of diets and corresponding orts were collected daily from d 20 to 28. Dry matter was determined by drying a portion of each sample in a forced-air oven at 105°C for 24 h.

Milk yields were recorded electronically on d 20 to 26 of each period (ProVantage Information Management System; Bou-Matic, Madison, WI). Each period, milk samples for each cow were collected from 6 consecutive milkings on d 25 and 26. Fat; true protein; anhydrous lactose; solids nonfat; urea N; de novo, mixed, preformed fatty acids; and unsaturated double bonds were determined by mid-infrared procedures (CombiScope FTIR 300 Hp, Delta Instruments, Drachten, The Netherlands). Milk samples were mathematically composited in proportion to milk yield at each sampling by day and averaged for the period after analysis. Energy-corrected milk was calculated using a formula modified to account for use of true protein instead of total protein (Tyrrell and Reid, 1965; Hall, 2023): 0.327 × kilograms of milk + 12.95 × kilograms of fat + 7.65 × kilograms of true protein.

Feed efficiency (kg/kg) was calculated and expressed as milk yield/DMI and ECM/DMI for d 20 through d 26 of each period. Body weight was measured (Allweigh computerized scale; Allweigh Scale System Inc., Red Deer, AB, Canada), and BCS was assigned in 0.25-unit increments on a 1 to 5 scale (Ferguson et al., 1994) for each cow the day before the start of the study and then on d 28 of each period by 2 individuals and the average was reported.

Chewing Behavior. Eating and ruminating activity were recorded every 5 min for 3 consecutive 24-h periods (d 23 to d 25) for each period. While cows were out of the tie stall for milking, behavior recording continued. By multiplying the total number of observations for each activity by 5 min, the total time in minutes for each activity for each day was calculated. Meal bouts were determined with a 20-min interbout criterion, with new bouts established if the cow spent greater than 20 min performing another behavior before performing the same behavior (Black et al., 2016). Number of meals and meal length were recorded.

Feed Ingredients and Diets. The DM of individual feed ingredients was determined by collecting samples 3 times weekly and drying in a forced-air oven at 105°C for 24 h. Diets were adjusted according to change in DM when a feed ingredient DM value was more than ±1.2 SD.

During the collection period, feed ingredients, diets, and corresponding orts were collected on d 20 to 28 and a portion of each sample was dried in a forced-air oven at 105°C for 24 h for DM determination. Equal volumes of each sample from each collection day were frozen at −20°C and then composited by period. Diets and corresponding orts for d 20 to 22 were composited for apparent total-tract nutrient digestibility measurements. Diets were composited by treatment by period and orts were composited by cow by period. Feed ingredients for d 23 to 28 were composited for dietary composition. Period composites were stored at −20°C before analyses. From each period composite, a portion of each sample

Table 1. Ingredient composition (% of DM) of diets differing in uNDF240 and peNDF fed to lactating dairy cows

	Low uNDF240		High uNDF240	
Ingredient	Low peNDF	High peNDF	Low peNDF	High peNDF
Corn silage	34.68	34.68	34.68	34.68
Long-chopped timothy hay	—	10.48	—	24.19
Short-chopped timothy hay	10.48	—	24.19	—
Wheat straw, chopped	1.61	1.61	1.61	1.61
Concentrate mix				
Steam-flaked corn	9.68	9.68	8.32	8.32
Fine corn meal	5.68	5.68	7.10	7.10
Aminomax Pro[1]	7.72	7.72	9.12	9.12
Soybean meal	5.71	5.71	5.71	5.71
Soybean hulls	3.23	3.23	—	—
Wheat middlings	—	—	1.61	1.61
Beet pulp pellets	12.90	12.90	0.36	0.36
Canola meal	1.42	1.42	—	—
Rumen inert fat[2]	2.01	2.01	2.01	2.01
PGI amino enhancer[1]	0.73	0.73	0.73	0.73
Sugar, 99%	—	—	0.45	0.45
Calcium carbonate	0.89	0.89	1.14	1.14
Sodium sesquicarbonate	0.74	0.74	0.74	0.74
Energizer Gold[3]	0.79	0.79	0.94	0.94
Salt	0.42	0.42	0.42	0.42
Magnesium oxide	0.32	0.32	0.32	0.32
Urea	0.25	0.25	—	—
Omnigen-AF[4]	0.18	0.18	0.18	0.18
Chromium propionate	0.04	0.04	0.04	0.04
Trace mineral and vitamin premix[5]	0.11	0.11	0.11	0.11
Meta Smart[6]	0.05	0.05	0.05	0.05
Smartamine M[6]	0.04	0.04	0.04	0.04
AjiPro-L Gen 2[7]	0.03	0.03	0.03	0.03
XPC yeast culture[8]	0.05	0.05	0.05	0.05
Avail-Zn 120[9]	0.03	0.03	0.03	0.03
Mono dicalcium phosphate	0.19	0.19	—	—
Rumensin[10]	0.01	0.01	0.01	0.01
Total	100.00	100.00	100.00	100.00

[1]Poulin Grain; Newport, VT.

[2]BergaFat; Berg + Schmidt America, LLC; Libertyville, IL.

[3]Commercial fat, Poulin Grain; Newport, VT.

[4]Phibro Animal Health Corp., Teaneck, NJ.

[5]Contained 21.66% Ca, 0.91% Cl, 0.72% Mg, 0.17% P, 0.16% S, 0.01% K, 25,438 mg/kg Zn, 21,802 mg/kg Mn, 6,427 mg/kg Cu, 500 mg/kg Fe, 428 mg/kg I, 269 mg/kg Se, 154 mg/kg Co, 5,732 kIU/kg vitamin A, 1,589 kIU/kg vitamin D, and 29,762 kIU/kg vitamin E.

[6]Adisseo USA Inc.; Alpharetta, GA.

[7]Ajinomoto Heartland Inc., Chicago, IL.

[8]Diamond V Mills Inc.; Cedar Rapids, IA.

[9]Zinpro Corp., Eden Prairie, MN.

[10]Elanco Animal Health; Greenfield, IN.

was analyzed for chemical composition (CPM Plus; Cumberland Valley Analytical Services Inc., Waynesboro, PA). Analyses included DM, ash (method 942.05; AOAC International, 2012), CP (method 990.03; AOAC International, 2012), soluble CP (Krishnamoorthy et al., 1982), crude fat (method 2003.05; AOAC International, 2012), ADF (method 973.18; AOAC International, 2012), NDF using α-amylase and expressed on an OM basis (**aNDFom**; Van Soest et al., 1991), ADL (Goering and Van Soest, 1970), starch (Hall, 2009), sugars as ethanol soluble carbohydrates (DuBois et al., 1956), and minerals (method 985.01; AOAC International, 2012). Compositional information for the feed ingredients is summarized in Table 2.

Particle size distribution on a DM basis (55°C) was determined for each period composite sample by dry vertical sieving (Ro-Tap testing sieve shaker model B; W. S. Tyler Combustion Engineering Inc., Mentor, OH). Physically effective NDF of the treatment diets was calculated as the product of NDF content and pef using the standard method defined by Mertens (1997). Particle size information for feed ingredients is summarized in Table 3.

Using an in vitro rumen fermentation system (Raffrenato et al., 2018), uNDF on an OM basis was de-

Table 2. Analyzed chemical composition (mean ± SD), in vitro NDF degradability, and fermentation analysis of ingredients used for experimental diets (% of DM unless otherwise indicated)

Item	Corn silage	Long-chopped timothy hay	Short-chopped timothy hay	Wheat straw, chopped	Beet pulp pellets	Low uNDF240[1] grain mix	High uNDF240 grain mix
DM,[2] %	33.8 ± 2.3	88.3 ± 1.1	88.6 ± 0.9	88.2 ± 1.0	89.4 ± 0.5	87.7 ± 0.2	87.5 ± 0.2
CP	7.6 ± 0.4	11.2 ± 0.5	11.9 ± 0.6	3.2 ± 0.5	8.1 ± 0.5	25.6 ± 1.3	26.0 ± 0.8
Soluble protein, % of CP	4.8 ± 0.3	4.1 ± 0.4	4.1 ± 0.3	1.2 ± 0.3	1.1 ± 0.4	6.1 ± 1.4	4.2 ± 0.6
ADF	24.3 ± 1.4	40.9 ± 1.3	40.7 ± 0.3	54.0 ± 1.2	32.4 ± 4.7	10.6 ± 0.3	7.5 ± 0.7
aNDFom[3]	39.2 ± 3.0	66.2 ± 2.0	64.6 ± 1.6	81.9 ± 0.9	38.4 ± 2.3	16.1 ± 0.9	13.0 ± 1.3
ADL	3.2 ± 0.2	6.6 ± 0.2	6.4 ± 0.1	9.0 ± 0.7	4.5 ± 2.4	3.0 ± 0.0	2.8 ± 0.1
Starch	35.1 ± 1.7	0.7 ± 0.2	0.5 ± 0.1	1.0 ± 0.4	0.7 ± 0.3	30.4 ± 1.6	28.8 ± 0.9
Sugar (ESC[4])	0.9 ± 0.5	7.2 ± 0.1	7.5 ± 0.2	1.0 ± 0.3	10.0 ± 3.6	4.7 ± 0.4	5.3 ± 0.3
Ether extract	3.3 ± 0.1	1.9 ± 0.3	2.1 ± 0.1	1.4 ± 0.2	1.0 ± 0.2	4.6 ± 0.5	4.8 ± 0.8
Ash	3.3 ± 0.3	7.5 ± 0.4	8.2 ± 0.5	4.7 ± 0.7	12.5 ± 2.9	9.8 ± 1.1	11.1 ± 0.2
Calcium	0.26 ± 0.03	0.34 ± 0.04	0.39 ± 0.03	0.28 ± 0.03	1.53 ± 0.21	1.78 ± 0.19	1.98 ± 0.07
Phosphorus	0.22 ± 0.01	0.24 ± 0.01	0.26 ± 0.01	0.07 ± 0.06	0.09 ± 0.01	0.61 ± 0.03	0.59 ± 0.02
Magnesium	0.19 ± 0.03	0.20 ± 0.02	0.21 ± 0.00	0.10 ± 0.02	0.39 ± 0.05	0.77 ± 0.04	0.78 ± 0.03
Potassium	0.96 ± 0.10	2.15 ± 0.07	2.23 ± 0.09	1.03 ± 0.07	0.38 ± 0.06	1.12 ± 0.05	1.22 ± 0.01
Sulfur	0.14 ± 0.01	0.19 ± 0.01	0.20 ± 0.01	0.09 ± 0.02	0.38 ± 0.11	0.39 ± 0.03	0.44 ± 0.05
Sodium	0.01 ± 0.01	0.04 ± 0.01	0.04 ± 0.00	0.01 ± 0.01	0.12 ± 0.05	1.16 ± 0.18	1.20 ± 0.07
Chloride ion	0.22 ± 0.01	0.82 ± 0.02	0.82 ± 0.02	0.16 ± 0.07	0.04 ± 0.03	0.72 ± 0.10	0.81 ± 0.01
Iron, mg/kg	197 ± 20	220 ± 50	286 ± 111	117 ± 37	1,877 ± 345	453 ± 29	344 ± 41
Copper, mg/kg	7 ± 1	8 ± 0	8 ± 0	4 ± 1	9 ± 1	24 ± 2	24 ± 2
Manganese, mg/kg	22 ± 2	36 ± 3	38 ± 4	38 ± 18	120 ± 20	84 ± 3	86 ± 6
Zinc, mg/kg	26 ± 1	34 ± 2	35 ± 1	9 ± 6	28 ± 5	199 ± 13	205 ± 4
Lactic acid	4.6 ± 0.6	—	—	—	—	—	—
Acetic acid	4.3 ± 0.7	—	—	—	—	—	—
Propionic acid	0.6 ± 0.2	—	—	—	—	—	—
Isobutyric acid	—	—	—	—	—	—	—
Butyric acid	—	—	—	—	—	—	—
Total VFA	8.9 ± 0.9	—	—	—	—	—	—
pH	4.0 ± 0.1	—	—	—	—	—	—
uNDFom[5]							
12 h	—	—	—	—	16.7 ± 5.5	12.4 ± 1.7	11.1 ± 1.9
30 h	20.6 ± 2.7	39.9 ± 5.6	39.7 ± 3.3	57.1 ± 3.6	—	—	—
72 h	—	—	—	—	8.4 ± 3.7	3.5 ± 1.2	4.4 ± 1.6
120 h	10.8 ± 1.2	25.1 ± 2.2	22.8 ± 1.3	32.9 ± 2.3	7.0 ± 4.0	3.5 ± 1.4	4.7 ± 1.6
240 h	10.7 ± 0.9	22.9 ± 2.0	21.8 ± 1.1	31.0 ± 2.0	—	—	—

[1]uNDF240 = undegraded NDF at 240 h of in vitro fermentation.

[2]Ingredients were analyzed for DM at 105°C; n = 32; for all other analyses, n = 4.

[3]Amylase- and sodium sulfite-treated NDF, ash-corrected.

[4]Ethanol soluble carbohydrates.

[5]Undegraded NDF at the specified hour of in vitro fermentation, ash-corrected.

termined for 30-, 120- and 240-h time points for composite samples. The uNDFom for 12-, 72-, and 120-h time points for grains and nonforage fiber sources was determined according to Zontini (2016). Fermentation analyses were performed on the ensiled forage composite samples (Cumberland Valley Analytical Services Inc., Waynesboro, PA).

Rumen Evacuations and Analysis of Pool Size and Turnover. Ruminal contents of the cannulated cows were evacuated manually through the ruminal cannula after daily feeding on d 27 and before the end of d 28. To ensure that cows experienced the same interval of time between rumen evacuations, cows were divided into 2 groups of 4 cows each. The first group was evacuated 3.5 h after feeding on d 27 (1730 h) and 3.5 h before feeding of d 1 of next treatment period (1030 h). The second group of cows were evacuated 4.5 h after feeding on d 27 (1830 h) and 4.5 h before feeding on d 1 of next treatment (0930 h). Mass and volume of ruminal contents were determined. While evacuating, hand grab samples representing 10% of the contents were subsampled into a separate container. Subsamples of solid and liquid phases were separated using a nylon screen (1-mm pore size) and each weighed. Aliquots (approximately 300 g) from both the solid and liquid phases were collected and stored frozen at −20°C until processing. Within 30 min of initiating the evacuation, the remaining ruminal contents were returned through the ruminal cannula. Solid-phase aliquots were dried using a forced-air oven at 55°C for 24 h and ground through a 1-mm screen (Wiley mill; Arthur H. Thomas, Philadelphia, PA). Liquid-phase aliquots were dried using a forced-air oven at 55°C for 48 h and ground through a 2-mm screen (UDY Cyclone Sample Mill;

Table 3. Particle size characteristics of the ingredients used for the experimental diets (% of DM unless otherwise indicated)[1]

Item	Corn silage	Long-chopped timothy hay	Short-chopped timothy hay	Wheat straw, chopped	Beet pulp pellets	Low uNDF240[2] grain mix	High uNDF240 grain mix
Particle size distribution[3]							
>19.00 mm	0.5 ± 0.1	0.1 ± 0.0	0.0 ± 0.0	0.3 ± 0.2	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
13.20–19.00 mm	0.6 ± 0.2	0.4 ± 0.1	0.0 ± 0.0	0.9 ± 0.3	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
9.50–13.20 mm	4.3 ± 0.6	1.3 ± 0.0	0.1 ± 0.1	2.1 ± 0.3	1.5 ± 1.5	0.4 ± 0.3	0.2 ± 0.1
6.70–9.50 mm	14.7 ± 0.3	3.4 ± 0.5	0.1 ± 0.0	5.2 ± 1.0	86.9 ± 5.2	5.6 ± 0.6	4.7 ± 0.4
4.75–6.70 mm	20.8 ± 0.7	4.3 ± 0.3	0.3 ± 0.0	6.0 ± 0.7	2.3 ± 0.7	5.6 ± 0.4	5.6 ± 0.5
3.35–4.75 mm	21.0 ± 0.8	8.0 ± 0.7	0.9 ± 0.2	12.5 ± 1.7	1.4 ± 0.6	4.1 ± 0.3	3.8 ± 0.4
2.36–3.35 mm	14.6 ± 0.4	12.7 ± 0.7	3.3 ± 0.4	15.5 ± 1.6	0.9 ± 0.5	3.2 ± 0.1	3.5 ± 0.2
1.18–2.36 mm	13.9 ± 0.3	34.3 ± 2.7	34.0 ± 6.8	33.7 ± 0.8	1.8 ± 1.0	16.4 ± 0.3	17.7 ± 0.8
0.60–1.18 mm	5.4 ± 0.1	23.3 ± 3.0	53.8 ± 13.5	15.7 ± 1.4	1.5 ± 0.9	22.6 ± 0.4	23.4 ± 0.3
0.30–0.60 mm	2.5 ± 0.1	13.6 ± 3.5	41.9 ± 10.9	4.4 ± 0.9	1.5 ± 0.9	20.9 ± 0.6	19.9 ± 1.0
<0.30 mm	1.8 ± 0.1	8.8 ± 3.5	34.2 ± 10.7	1.6 ± 0.6	2.2 ± 1.5	21.2 ± 0.8	21.2 ± 0.9
peNDF[4]	35.4 ± 1.2	37.6 ± 5.1	15.3 ± 1.8	62.9 ± 5.2	40.9 ± 3.1	5.9 ± 0.6	4.9 ± 0.4

[1]Mean ± SD; n = 4 samples/ingredient.

[2]uNDF240 = undegraded NDF at 240 h of in vitro fermentation.

[3]Measured with dry vertical sieving using a Ro-Tap shaker.

[4]peNDF = physically effective NDF, (% of sample DM retained on the 1.18-mm screen and greater) × (NDF content of ingredient).

UDY Corp., Fort Collins, CO). Corresponding solid and liquid phases were recombined based on the proportion of DM. Recombined ruminal contents were analyzed for ash (modified method 942.05; AOAC International, 2012; 4 h at 600°C), aNDFom (as described previously), uNDF240om (as described previously), and starch (Cumberland Valley Analytical Services Inc., Waynesboro, PA).

Ruminal pool size of OM, NDF, uNDF240om, and starch were calculated as the product of DM mass of ruminal contents and nutrient content. Ruminal turnover rate (%/h) of OM, NDF, uNDF240om, and starch were calculated as [100 × (intake of nutrient/ruminal pool of nutrient)/24], according to Oba and Allen (2000). Nutrient intake was calculated using DMI from d 27 and 28 and the nutrient content of the diets from d 23 to 28. Ruminal turnover time (hours) was calculated as 1/[ruminal turnover rate (%/h)/100].

Ruminal Fermentation. Indwelling ruminal pH meters (Penner et al., 2006; LRCpH; Dascor, Escondido, CA) were used to measure ruminal pH at 30-s intervals for 96 consecutive h on d 23 to 26 in the 8 ruminally fistulated cows. The electrodes were standardized at the start and finish of the 96-h measurement period and the shift in millivolt readings was assumed to be linear (Penner et al., 2007). Ruminal pH was averaged over a 10-min period. Mean pH, minimum pH, maximum pH, the area below a pH of 5.8, and hours per day that pH was below 5.5 or 5.8 as an indicator of subacute ruminal acidosis, were calculated using 10-min-period ruminal pH values.

Ruminal fluid samples (approximately 500 mL), hand grabbed from below the ruminal digesta mat, were collected each period on d 26 every 4 h for 24 h according to the protocol of Miller et al. (2020). After straining through 4 layers of cheesecloth, a portion (approximately 40 mL) of the ruminal fluid was separated for analysis for VFA concentration (Supelco Inc., 1998) by GC with use of a Varian CP-3800 gas chromatograph (Varian Inc., Palo Alto, CA) equipped with a flame-ionization detector and an 80/120 Carbopack B-DA/4% Carbowax 20M column (Supelco Inc., Bellefonte, PA). Additionally, 10 mL of ruminal fluid were added to 100 µL of concentrated HCl for analysis of ruminal NH_3-N concentration. The NH_3-N concentration was determined according to the procedure of Chaney and Marbach (1962). Both ruminal fluid aliquots were stored at −20°C before analysis.

Apparent Total-Tract Nutrient Digestibility. Apparent total-tract digestibility of OM, aNDFom, pdNDF, and starch was determined on d 20 to 22 of each period. As previously described, diets and corresponding orts were composited accordingly for d 20 to 22. Fecal grab samples were collected starting at d 20 at 1430 h (beginning of study day) then every 9 h for a total of 8 samples to represent every 3 h in a 24-h period for each period. The specific hours for spot fecal sampling were: d 20 at 1500, 2400, and 0900 h; d 21 at 1800, 0300, and 1200 h; and d 22 at 2100 and 0600 h. The technical start of the study day was at 1430 h with feeding. Previous research has found that a sampling frequency of at least 6 times daily with lactating dairy cattle is necessary to accurately estimate nutrient digestibility with indigestible NDF as a marker in lieu of total fecal collection (Morris et al., 2018). Approximately 100 g of feces from each time point were combined by cow for each period composite. Diet, orts, and fecal samples were dried in a forced-air oven at 55°C for 48 h before being ground to pass through a 1-mm screen (Wiley mill; Arthur H. Thomas, Philadelphia, PA). Total-tract composite samples of diets (by period), orts (by cow), and feces (by cow) were

Table 4. Composition (mean ± SD) of experimental diets differing in uNDF240 and peNDF based on chemical analysis of individual ingredients (% of DM unless otherwise indicated)

	Low uNDF240		High uNDF240	
Item	Low peNDF	High peNDF	Low peNDF	High peNDF
DM,[1]%	57.9 ± 2.2	57.6 ± 1.9	58.2 ± 1.9	57.5 ± 1.9
CP	15.3 ± 0.6	15.2 ± 0.6	15.7 ± 0.3	15.5 ± 0.3
Soluble protein, % of CP	37.2 ± 2.3	37.4 ± 2.4	36.9 ± 1.1	37.4 ± 1.7
ADF	22.0 ± 0.3	22.0 ± 0.2	22.2 ± 0.6	22.2 ± 0.9
aNDFom[2]	33.1 ± 1.7	33.3 ± 1.8	35.7 ± 1.8	36.1 ± 2.0
ADL	3.7 ± 0.2	3.7 ± 0.3	3.9 ± 0.1	3.9 ± 0.0
Starch	24.6 ± 0.6	24.6 ± 0.6	23.4 ± 0.9	23.5 ± 0.9
Sugar	4.3 ± 0.5	4.3 ± 0.5	4.6 ± 0.3	4.6 ± 0.2
Ether extract	3.4 ± 0.2	3.4 ± 0.2	3.6 ± 0.3	3.5 ± 0.4
Ash	7.6 ± 0.7	7.6 ± 0.7	7.5 ± 0.2	7.4 ± 0.2
Calcium	1.05 ± 0.11	1.05 ± 0.11	0.96 ± 0.04	0.95 ± 0.04
Phosphorus	0.36 ± 0.01	0.3 6 ± 0.01	0.37 ± 0.01	0.37 ± 0.01
Magnesium	0.45 ± 0.02	0.45 ± 0.02	0.4 2 ± 0.02	0.42 ± 0.02
Potassium	0.36 ± 0.04	0.36 ± 0.04	0.37 ± 0.05	0.37 ± 0.04
Sulfur	0.28 ± 0.02	0.28 ± 0.02	0.27 ± 0.02	0.27 ± 0.02
Sodium	0.49 ± 0.07	0.49 ± 0.07	0.48 ± 0.03	0.48 ± 0.03
Chloride ion	0.46 ± 0.04	0.46 ± 0.04	0.59 ± 0.00	0.59 ± 0.00
Iron, mg/kg	525 ± 49	518 ± 46	279 ± 26	263 ± 14
Copper, mg/kg	14 ± 1	14 ± 1	14 ± 1	14 ± 1
Manganese, mg/kg	61 ± 3	61 ± 3	51 ± 3	51 ± 2
Zinc, mg/kg	97 ± 5	96 ± 6	97 ± 2	97 ± 2
uNDFom[3]				
30 h	19.4 ± 0.7	19.4 ± 0.9	22.0 ± 0.7	22.1 ± 1.1
120 h	9.2 ± 0.3	9.4 ± 0.2	11.5 ± 1.5	12.1 ± 2.1
240 h	8.8 ± 0.5	8.9 ± 0.5	11.4 ± 1.2	11.6 ± 1.1

[1]Diets were analyzed for DM at 105°C; n = 32 for each diet; for all other analyses, n = 4.

[2]Amylase- and sodium sulfite-treated NDF, ash-corrected.

[3]Undegraded NDF at the specified hour of in vitro fermentation, ash-corrected.

submitted for wet chemistry analysis (Cumberland Valley Analytical Services Inc., Waynesboro, PA) for DM, ash, aNDFom, starch, and uNDF240om as previously described. Sample uNDF240om was used as an internal marker. Apparent total-tract digestibility was calculated by the ratio technique using the concentrations of the nutrients and uNDF240om in the diet and feces. The nutrient content of the diet used in the digestibility calculation was adjusted for each cow based on the nutrient composition of the diet offered and refused.

Statistical Analysis

Data from the analysis of feed ingredients and diets were analyzed using the MEANS procedure of SAS (Statistical Analysis System, version 9.4; SAS Institute Inc., Cary, NC) and are reported as descriptive statistics (mean ± SD).

One cow did not finish the study due to a severe case of mastitis (*Klebsiella*). A second cow was removed from the study due to severe mastitis (*Staphylococcus* spp). The second animal responded to treatment and finished the study but was excluded from the dataset due to milk yield being less than 50% of production before the occurrence of mastitis. All data from the 2 cows were completely removed from the dataset. Thus, 14 cows (8 cannulated and 6 noncannulated) were used for statistical analysis.

Data were checked for homogeneity of variance and normality assumptions using Shapiro–Wilk and Levene's tests using the GLM procedure of SAS. Data for DMI, milk yield and composition, feed efficiency, apparent total-tract nutrient digestibility, passage rates, BW, and BCS were analyzed according to the following model using the MIXED procedure of SAS:

$$Y_{ij(mn)} = \mu + P_i + K_j + A_m + B_n + (AB)_{mn} + e_{ij(mn)},$$

where $Y_{ij(mn)}$ is the dependent variable, μ is the overall mean, P_i is the random effect of cow within square (i = 1–4), K_j is the fixed effect of period (j = 1–4), A_n is the main effect of uNDF240 (m = 1–2), B_n is the main effect of peNDF at the nth level (n = 1–2), $(AB)_{mn}$ is the interaction effect between uNDF240 and peNDF, and $e_{ij(mn)}$ is the residual error.

Repeated measurements on performance data (i.e., DMI, milk yield, and milk composition) were reduced to period means for each cow before statistical analysis.

Ruminal starch turnover failed normality assumptions and was log-transformed. Least squares means and 95% confidence limits are presented as nontransformed values,

Table 5. Particle size characteristics of experimental diets differing in uNDF240 and peNDF (all values % of DM unless otherwise indicated)[1]

	Low uNDF240		High uNDF240	
Item	Low peNDF	High peNDF	Low peNDF	High peNDF
Particle size distribution[2]				
>19.00 mm	0.2 ± 0.4	0.0 ± 0.1	0.0 ± 0.0	0.1 ± 0.1
13.20–19.00 mm	0.1 ± 0.1	0.2 ± 0.1	0.1 ± 0.1	0.2 ± 0.2
9.50–13.20 mm	1.0 ± 0.2	1.3 ± 0.3	1.3 ± 0.5	2.2 ± 0.8
6.70–9.50 mm	14.2 ± 2.8	15.3 ± 3.2	6.1 ± 1.1	8.4 ± 0.9
4.75–6.70 mm	10.2 ± 0.9	10.3 ± 0.6	8.2 ± 0.3	9.6 ± 0.6
3.35–4.75 mm	10.3 ± 0.8	11.3 ± 1.1	9.0 ± 0.3	10.3 ± 1.5
2.36–3.35 mm	8.1 ± 0.5	9.3 ± 0.8	7.6 ± 0.7	9.9 ± 0.7
1.18–2.36 mm	16.7 ± 1.1	17.9 ± 1.4	19.7 ± 0.9	20.2 ± 1.2
0.60–1.18 mm	17.0 ± 1.0	15.3 ± 1.5	21.6 ± 1.2	18.1 ± 1.8
0.30–0.60 mm	13.2 ± 0.4	11.2 ± 0.6	15.6 ± 0.9	12.2 ± 1.0
<0.30 mm	9.0 ± 1.0	7.9 ± 0.9	10.9 ± 0.9	8.9 ± 0.8
Physical effectiveness factor (pef)[3]	0.608 ± 0.009	0.656 ± 0.022	0.520 ± 0.016	0.608 ± 0.026
peNDF[4]	20.1 ± 1.3	21.8 ± 1.7	18.6 ± 1.4	22.0 ± 2.1
peuNDF240[5]	5.4 ± 0.1	5.8 ± 0.1	5.9 ± 0.1	7.1 ± 0.3

[1]Mean ± SD; n = 4 samples/ingredient.

[2]Measured with dry vertical sieving using a Ro-Tap shaker.

[3]Fraction of sample DM retained on the 1.18-mm screen and greater.

[4]peNDF = pef × aNDFom.

[5]Physically effective undegraded NDF after 240 h of in vitro fermentation, ash-corrected; pef × uNDF240.

and *P*-values refer to the model effects corresponding to the transformed data. Significance was declared at $P \leq 0.05$ and trends at $0.05 < P \leq 0.10$.

RESULTS AND DISCUSSION

Environmental Conditions

The average temperature during the study was 19.0°C (±0.04°C) and relative humidity was 79.4% (±0.1%). These measurements were below the heat stress threshold and within the thermoneutral zone of lactating dairy cattle (Atrian and Shahryar, 2012).

Ingredient and Dietary Nutrient Composition

Dietary aNDFom content ranged from 33.2% for LU diets to 35.9% of dietary DM for the HU diets and these concentrations were expected to be and similar to previous studies with corn silage–based diets (Oba and Allen, 2000; Miller et al., 2021; Vieira et al., 2025). Increasing dietary forage from 46.8% to 60.5% of DM boosted dietary uNDF240om content (% of DM) from approximately 8.8% (LU-LP and LU-HP) to 11.5% (HU-LP and HU-HP; Table 4). These uNDF240 values span a range from moderate to high concentration based on the reported range of 6% to 11% of DM reported by Cotanch et al. (2014) for lactating dairy cows fed both corn silage–based and alfalfa-based diets. Although there has been relatively little published research examining the effect of uNDF240om on DMI, we expected that 8.8% uNDF240om in a corn silage–based diet would pose no constraint to DMI, whereas 11.5% uNDF240om would limit DMI as reported for both low-digestibility alfalfa and corn silage–based diets (Fustini et al., 2017; Miller et al., 2021).

Dietary peNDF concentrations were approximately 19% (LU-LP and HU-HP) or 22% of DM (LU-HP and HU-HP; Table 5). Within each level of uNDF240om, dietary peNDF content spanned the commonly cited requirement of approximately 21% for lactating cows and the peNDF relationships reported by Mertens (1997) would predict biologically meaningful differences in milk and ruminal responses. The lower and higher peNDF diets also differed substantially in particle size distribution. Specifically, pef was approximately 12% less for LP than HP diets, and HP diets contained approximately 5 to 9 percentage units more physically effective particles retained on the ≥1.18-mm sieve (Table 5). Consequently, we concluded that there was sufficient spread in dietary particle size and undegradable fiber to successfully test our hypotheses.

To quantitate the physically effective portion of uNDF240om, peuNDF240 was calculated for each diet as pef × uNDF240om (Table 5). We intended this metric to integrate the effects of particle size and NDF undegradability into one number. Dietary peuNDF240 concentrations were 5.4% (LU-LP), 5.8% (LU-HP), 5.9% (HU-LP), and 7.1% of ration DM (HU-HP; Table 5). Using these values, we aimed to assess the effects of low and high dietary peuNDF240 concentrations, and to determine if the 2 intermediate peuNDF240 diets (LU-HP and HU-LP) elicited numerically similar responses to diet despite differing in particle size and fiber undegradability.

Aside from differences in uNDF240om, peNDF, and peuNDF240, the diets contained similar contents of all other measured nutrients and these nutrients were within recommended ranges (NASEM, 2021; Table 4).

Dry Matter and Fiber Intake

The interaction of dietary uNDF240om and peNDF was significant ($P = 0.02$) for DMI, which has considerable practical importance for ration formulation (Table 6). With the 8.8% uNDF240om diets, particle size had no effect on DMI. However, for the 11.5% uNDF240om diets, which were expected to constrain DMI via gut fill (based on Miller et al., 2021), smaller particle size released the DMI constraint allowing DMI to increase by 2.5 kg/d over the diets with larger particle size (Table 6). In fact, reducing the particle size of the HU-HP diet resulted in DMI similar to the LU diets. Previous studies have observed variable responses in DMI to particle size, although it appears that a reduction in forage particle size has a greater effect on DMI for higher forage diets that are also higher in uNDF240om, as in our study (Beauchemin et al., 2003; Teimouri Yansari et al., 2004; Farmer et al., 2014; Haselmann et al., 2019). As an example, Haselmann et al. (2019) fed dairy cows a high-forage diet (grass hay and clover-grass silage) and found that finer chopped forages increased DMI by 1.8 kg/d. Consequently, they recommended reducing forage particle size to improve DM and energy intake when cows are fed higher forage diets.

Indigestible NDF has been proposed as a primary constituent of forage NDF that limits DMI, especially as forage content in the diet increases (Harper and McNeill, 2015). Miller et al. (2021) concluded that, for corn silage–based diets, gut fill constraints on DMI become apparent at about 10% uNDF240om (percent of ration DM). Using 288 h of in situ fermentation as their measure of uNDF, Vieira et al. (2025) found that DMI for corn silage–based diets decreased as uNDF increased between 9.0% and 11.1% of ration DM. Similarly, we observed a reduction in DMI as dietary uNDF240om increased from 8.8% to 11.5% of DM, but only with greater particle size. Other uNDF time points may be usefully related to DMI, alone or in combination with uNDF240, that reflect the relative proportions of faster and slower degrading NDF (e.g., 30-h uNDF; Kahyani et al., 2019). Tables 2 and 4 contain uNDF values measured at 12, 30, 72, 120, and 240 h of in vitro fermentation so that any combination of uNDF values may be considered.

As with DMI, there was a significant interaction between dietary uNDF240om and peNDF for aNDFom intake ($P = 0.03$) and uNDF240om intake ($P = 0.007$; Table 6). This interaction reflects the greater DMI for the HU-LP diet, which translated to 0.8 kg/d more aNDFom

Table 6. Least squares means of intake and BW data for lactating Holstein cows (n = 14) fed diets differing in uNDF240 and peNDF

	Low uNDF240		High uNDF240			*P*-value		
Item	Low peNDF	High peNDF	Low peNDF	High peNDF	SE	uNDF	peNDF	uNDF × peNDF
DMI, kg/d	27.5	27.3	27.4	24.9	0.6	0.01	0.005	0.02
DMI, % of BW/d	4.02	4.04	3.99	3.73	0.10	0.01	0.05	0.03
aNDFom intake,[1] kg/d	9.12	9.06	9.74	8.96	0.19	0.11	0.01	0.03
aNDFom intake, % of BW/d	1.33	1.34	1.42	1.34	0.04	0.04	0.10	0.05
peNDF intake, kg/d	5.56	5.94	5.07	5.44	0.11	<0.001	<0.001	0.95
peNDF intake, % of BW/d	0.81	0.88	0.74	0.81	0.02	<0.001	<0.001	0.82
uNDF240om intake,[2] kg/d	2.41	2.43	3.11	2.87	0.05	<0.001	0.02	0.007
uNDF240om intake, % of BW/d	0.35	0.36	0.45	0.43	0.01	<0.001	0.20	0.008
peuNDF240 intake,[3] kg/d	1.47	1.59	1.61	1.74	0.03	<0.001	<0.001	0.85
peuNDF240 intake, % of BW/d	0.22	0.24	0.24	0.26	0.01	<0.001	<0.001	0.54
BW, kg	689	682	688	675	20	0.11	<0.001	0.24
BCS	2.84	2.79	2.81	2.77	0.06	0.43	0.16	0.87

[1]Amylase- and sodium sulfite-treated NDF, ash-corrected.

[2]Undegraded NDF after 240 h of in vitro fermentation, ash-corrected.

[3]Physically effective undegraded NDF after 240 h of in vitro fermentation, ash-corrected.

Table 7. Least squares means of lactation performance data for lactating Holstein cows (n = 14) fed diets differing in uNDF240 and peNDF

	Low uNDF240		High uNDF240			*P*-value		
Item	Low peNDF	High peNDF	Low peNDF	High peNDF	SE	uNDF	peNDF	uNDF × peNDF
Milk, kg/d	46.0	44.5	43.9	42.2	0.9	<0.001	0.005	0.86
ECM, kg/d	47.5	45.2	46.5	44.1	1.0	0.12	0.002	0.96
Fat, %	3.68	3.66	3.93	3.92	0.10	<0.001	0.76	0.89
Fat, kg/d	1.70	1.62	1.71	1.64	0.05	0.53	0.02	0.93
True protein, %	2.93	2.88	2.96	2.84	0.06	0.83	<0.001	0.09
True protein, kg/d	1.35	1.27	1.29	1.19	0.03	<0.001	<0.001	0.56
Lactose (anhydrous), %	4.63	4.61	4.58	4.58	0.02	0.003	0.38	0.20
Lactose (anhydrous), kg/d	2.16	2.05	2.02	1.93	0.05	<0.001	0.004	0.79
Solids nonfat, %	8.63	8.56	8.61	8.49	0.06	0.10	<0.001	0.30
Solids nonfat, kg/d	4.01	3.80	3.78	3.56	0.08	<0.001	<0.001	0.94
Urea nitrogen, mg/dL	8.53	9.40	10.12	11.00	0.55	<0.001	0.03	1.00
De novo FA,[1] g/100 g milk	0.87	0.85	0.90	0.88	0.03	0.07	0.22	0.71
Mixed-origin FA, g/100 g milk	1.41	1.40	1.51	1.51	0.04	<0.001	0.82	0.97
Preformed FA, g/100 g milk	1.24	1.24	1.34	1.38	0.04	<0.001	0.30	0.42
Unsaturation, double bonds/FA	0.27	0.27	0.26	0.26	0.006	0.10	0.72	0.39
Milk/DMI, kg/kg	1.68	1.65	1.61	1.71	0.04	1.00	0.18	0.03
ECM/DMI, kg/kg	1.71	1.68	1.70	1.79	0.04	0.05	0.25	0.03

[1]FA = fatty acids.

and 0.2 kg/d more uNDF240om intake relative to the HU-HP diet. In contrast, particle size had little effect on aNDFom or uNDF240om intake with the LU diets (Table 6). Cows fed the HU-LP diet consumed 1.42% of BW/d as aNDFom while cows consuming the other 3 diets consumed 1.34% of BW/d (Table 6). Using NDF as a measure of a diet's ruminal filling effect, Mertens (2009) concluded that approximately 1.25% of BW/d was the NDF intake capacity for lactating dairy cows. The high aNDFom intake for cows fed the HU-LP diet (i.e., 1.42% of BW/d) confirms the substantial effect that reducing forage particle size has on DMI and NDF intake for higher forage, higher uNDF240om diets. Likewise, the 0.45% of BW/d intake of uNDF240om observed for cows fed the HU-LP treatment is near the presumed maximum uNDF240om intake for lactating dairy cows (~0.40% of BW/d; Cotanch et al., 2014) and is comparable to observations by Fustini et al. (2017) for cows fed a highly degradable alfalfa hay with a high uNDF240om content (0.48% of BW/d) and Vieira et al. (2025) for cows fed corn silage–based diets (0.48% of BW/d).

There was no interaction ($P = 0.85$) between uNDF240om and peNDF for peuNDF240 intake (Table 6). As expected, cows fed the higher uNDF240om and peNDF diets consumed more peuNDF240 than cows fed the lower uNDF240om and peNDF diets ($P < 0.001$; Table 6). Importantly, the peuNDF240 intake for cows fed the LU-HP and HU-LP diets averaged 1.60 kg/d, although the composition of the peuNDF240 differed markedly between the 2 diets. Specifically, diet LU-HP contained less uNDF240om chopped more coarsely whereas diet HU-LP contained more uNDF240om chopped finer. The peuNDF240 content tracked numerically with DMI, which agrees with Miller et al. (2021), who also reported a close relationship between dietary peuNDF240 (measured using the same method) and DMI for cows fed rations containing approximately 50% or 65% forage (70%–80% corn silage and 20%–30% haycrop silage). Vieira et al. (2025) measured pef using the Penn State Particle Separator (8- and 19-mm sieve) rather than the standard dry, vertical sieving method and measured uNDF at 288 h of in situ fermentation rather than 240 h of in vitro fermentation, and despite using different methodology, they also found that DMI tracked with uNDF and peuNDF when cows were fed corn silage–based diets. Future research that encompasses a wide range of uNDF240 from low (i.e., ~6% of ration DM) to high (i.e., ~12% of ration DM) could better assess the nature of the interaction with particle size for low-uNDF240 diets that may be deficient in fiber.

Milk Yield, Composition, and Efficiency of Milk Production

There was no interaction between uNDF240om and peNDF for either milk ($P = 0.86$) or ECM yield ($P = 0.96$). Cows fed 8.8% uNDF240om diets produced significantly more milk than those fed 11.5% uNDF240om (45.3 vs. 43.1 kg/d; $P < 0.001$). Also feeding corn silage–based diets, Miller et al. (2021) observed less ECM production when dietary uNDF240om was approximately 10% of DM. Another study with corn silage observed a significant linear increase in milk and ECM yield as dietary uNDF decreased within the range of 9% to 11% of ration DM (Vieira et al., 2025). Regardless of dietary uNDF240om, however, higher peNDF reduced milk ($P = 0.005$) and

Table 8. Least squares means of chewing behavior data for lactating Holstein cows (n = 14) fed diets differing in uNDF240 and peNDF

	Low uNDF240		High uNDF240			*P*-value		
Item	Low peNDF	High peNDF	Low peNDF	High peNDF	SE	uNDF	peNDF	uNDF × peNDF
Eating time								
min/d	255	263	279	300	12	<0.001	0.03	0.28
min/kg of DMI	9.1	9.6	10.1	11.9	0.5	<0.001	<0.001	0.01
Rumination time								
min/d	523	527	532	545	16	0.14	0.37	0.60
min/kg of DMI	18.6	19.3	19.2	21.7	0.8	<0.001	<0.001	0.01
Total chewing time								
min/d	779	789	811	845	25	<0.001	0.05	0.29
min/kg of DMI	27.7	28.9	29.3	33.6	1.2	<0.001	<0.001	<0.001
Meal length, min/meal	27.7	32.8	32.6	37.7	2.5	<0.001	<0.001	0.99
Meal bout, bouts/d	11.3	10.5	10.7	10.0	0.5	0.09	0.01	0.93

ECM yield ($P = 0.002$). Milk yield tracked numerically with dietary peuNDF240; it was least for the highest peuNDF240 diet, similar for diets LU-HP and HU-LP that had the same peuNDF240 though differing uNDF240om and peNDF, and greatest for the lowest peuNDF240 diet. This pattern of milk production agrees with Serva et al. (2021) who observed a significant negative relationship between dietary peuNDF240 and milk yield and gross feed efficiency across 22 commercial dairy farms feeding alfalfa- and corn silage–based rations.

No interaction occurred between uNDF240om and peNDF for percentage or production of milk fat, lactose, or SNF ($P > 0.10$; Table 7). Cows fed the 8.8% uNDF240om diets produced milk with significantly less ($P < 0.001$) milk fat content than cows fed 11.5% uNDF240om (3.67% versus 3.93%). Output of milk fat was significantly reduced ($P = 0.02$) for cows fed approximately 19% versus 22% peNDF diets (1.63 versus 1.71 kg/d). This reduction in milk fat content and output with lower dietary uNDF240om and peNDF has been observed in previous studies (Mertens, 1997; Fustini et al., 2017; Miller et al., 2021). However, our results are unique in that we examined, within one study, the effects of particle size, uNDF240om, and their interaction on DMI and lactational performance.

There was a trend for an interaction between uNDF240om and peNDF ($P = 0.09$) for milk true protein percentage. Similar to DMI, milk protein percentage was affected to a greater extent by peNDF at higher versus lower dietary uNDF240om. Previous research has found that smaller forage particle size is associated with greater efficiency of microbial protein synthesis because of faster passage rate of digesta (Rode et al., 1985). The difference in milk protein percentage and output between the LU-LP and HU-HP diets also may have been related to greater microbial yield linked with increased fermentable fiber (Van Soest, 1994) and decreased uNDF240om (Vieira et al., 2025). Output of milk true protein was significantly enhanced ($P < 0.001$) by both lower dietary uNDF240om and lower peNDF ($P < 0.001$). The net impact of these dietary effects was that milk true protein production was lowest for cows fed the HU-HP diet, greatest for the LU-LP diet, and similar for the diets with the same peuNDF240 (Table 7).

Milk urea nitrogen content was significantly reduced ($P < 0.001$), with lower dietary uNDF240om and smaller particle size ($P = 0.03$). These results likely reflect the greater fermentability of the diets containing more pdNDF and larger surface area for microbial attachment. In fact, Hristov and Ropp (2003) reported that diets providing more ruminally fermentable fiber enhanced the transfer of ruminal ammonia to milk protein thereby reducing MUN. As with DMI, ECM, and milk true protein output, milk MUN tracked numerically with dietary peuNDF240 content.

The primary effect of diet on milk fatty acid profiles was the significant ($P < 0.001$) reduction in mixed-origin fatty acids and a trend ($P = 0.07$) for lower de novo milk fatty acids for cows fed the lower uNDF240om diets, and greater ($P < 0.001$) preformed milk fatty acids for those cows fed higher uNDF240om diets ($P < 0.001$; Table 7). Palmitic acid is the major component of the mixed fraction, and it is also the longest fatty acid produced in the chain elongation process of de novo fatty acid synthesis. Consequently, it is the first de novo fatty acid to decrease when trans fatty acid–induced milk fat depression occurs and explains the reduction in both mixed and de novo fatty acids (Bauman and Griinari, 2003). The actual reduction in mixed fatty acids was small, reflecting the modest although significant depression in milk fat content for cows fed the low uNDF240om diets. Lower milk de novo fatty acid content for cows fed lower uNDF240om diets agrees with the previously reported positive relationship between milk de novo fatty acids and milk fat percentage (Barbano et al., 2018; Matamoros et al., 2020). The relationship between preformed fatty

Table 9. Ruminal pH data for lactating Holstein cows (n = 8) fed diets differing in uNDF240 and peNDF

	Low uNDF240		High uNDF240			*P*-value		
Item	Low peNDF	High peNDF	Low peNDF	High peNDF	SE	uNDF	peNDF	uNDF × peNDF
Daily mean pH	6.11	6.17	6.22	6.24	0.05	0.006	0.24	0.50
Minimum pH	5.48	5.51	5.64	5.55	0.09	0.03	0.51	0.18
Maximum pH	6.66	6.73	6.70	6.75	0.04	0.19	0.008	0.70
SD of pH	0.29	0.29	0.24	0.28	0.02	0.08	0.20	0.15
Time pH <5.8, min/d	253	208	166	164	61	0.07	0.49	0.53
Time pH <5.5, min/d	71	71	40	31	22	0.03	0.77	0.78
AUC <5.8[1]	52.0	49.6	33.5	30.0	15.0	0.06	0.76	0.95

[1]AUC = area under the curve <5.8 = ruminal pH units below 5.8 by 10 min.

acid content of milk and milk fat percentage has been reported to be very weak and related to dietary preformed fatty acids and mobilized adipose tissue (Barbano et al., 2017). However, no change in BW or BCS resulted from dietary uNDF240om content, and predicted supply of long-chain fatty acids using values in CNCPS v. 6.5.5 were similar across diets. Overall, changes in mixed and de novo milk fatty acid profile explained the observed changes in milk fat percentage.

There was a significant interaction (P = 0.03) between dietary uNDF240om and peNDF for efficiency of ECM production (Table 7). Smaller dietary particle size had no effect on efficiency for the lower uNDF240om diets, but it reduced ECM/DMI for the high uNDF240om diets. This response in ECM/DMI was driven by the reduction in DMI for cows fed the HU-HP diet. This diet also supported the lowest BW (Table 6) and part of the gross efficiency response may have been driven by mobilization of body tissue; a longer term study would be required to better determine the reason for the observed changes in milk production efficiency.

Chewing Behavior and Meal Characteristics

No significant interaction between uNDF240om and peNDF occurred for eating or ruminating time (min/d), but a significant interaction did exist for eating (P = 0.01), ruminating (P = 0.01), and total chewing time (P < 0.001) expressed as minutes per kilogram of DMI (Table 8). Cows fed higher uNDF240om spent approximately 12% more time eating with 6% more total chewing time than cows fed lower uNDF240om diets. Likewise, higher peNDF diets elicited significantly more time eating (P = 0.03) and total chewing time (P = 0.05) than lower peNDF. The lack of response in daily rumination time to either uNDF240om or peNDF likely was due to cows eating and reducing dietary particles to a uniform size before swallowing as reported by Schadt et al. (2012) and Grant and Cotanch (2023). Consequently, when expressed as minutes per kilogram of DMI, the effect of greater dietary particle size on eating activity was larger for the 11.5% than the 8.8% uNDF240om diet. Similar to eating time, the greater ingestive chewing per kilogram of DMI was explained by cows chewing higher NDF, lower NDF degradability forages more before swallowing, and taking more time to do so, in an effort to achieve a relatively uniform particle size of the swallowed bolus (Schadt et al., 2012; Grant and Cotanch, 2023). Lower dietary peNDF elicited less rumination per kilogram of DMI likely because more fiber particles in the swallowed bolus were at or below the critical size for ruminal escape (Grant and Cotanch, 2023). Considerable research illustrates the positive relationship between dietary peNDF and rumination time per kilogram of DMI (Allen and Grant, 2000; Beauchemin et al., 2003; Teimouri Yansari et al., 2004). In our study, this relationship was mainly observed when dietary uNDF240om was 11.5% rather than 8.8% of ration DM. As discussed earlier, the effect of dietary peNDF on DMI and chewing responses in our study was more pronounced when dietary uNDF240om content was high enough to constrain DMI.

There was no significant interaction between uNDF240om and peNDF for either meal length (P = 0.99) or number of meals (P = 0.93; Table 8). However, greater dietary uNDF240om lengthened meals by approximately 25% (P < 0.001) and tended to decrease (P = 0.09) meal bouts by about 0.6 meal/d. Greater peNDF increased meal length (P < 0.001) and decreased meal bouts (P = 0.01) for both lower and higher uNDF240om diets. These changes in meal patterns should result in a more stable ruminal pH for fiber fermentation (Pitt and Pell, 1997). As with DMI and ECM, meal length and number of daily meals mirrored the peuNDF240 content of the diet. By decreasing dietary uNDF240om and peNDF, ultimately decreasing peuNDF240, meal length decreased and frequency increased. As with ECM, the relative similarity in meal characteristics between the LU-HP and HU-LP diets indicates the potential of peuNDF240 to integrate the effects of fiber undegradability and particle size.

Table 10. Ruminal fermentation data for lactating Holstein cows (n = 8) fed diets differing in uNDF240 and peNDF

Item	Low uNDF240		High uNDF240		SE	*P*-value		
	Low peNDF	High peNDF	Low peNDF	High peNDF		uNDF	peNDF	uNDF × peNDF
Total VFA, m*M*	122.8	120.6	118.3	112.3	4.1	0.02	0.14	0.47
VFA, % of total VFA								
Acetate (A)	63.4	63.8	63.9	64.1	0.9	0.08	0.20	0.64
Propionate (P)	22.7	22.5	21.5	21.6	0.8	<0.001	0.81	0.41
Butyrate (B)	11.2	11.0	11.5	11.3	0.4	0.01	0.03	0.97
Isobutyrate	0.57	0.59	0.68	0.71	0.03	<0.001	0.18	0.67
Valerate	1.68	1.64	1.80	1.66	0.12	0.12	0.06	0.30
Isovalerate	0.45	0.51	0.62	0.62	0.03	<0.001	0.08	0.13
A:P	2.83	2.89	3.04	3.01	0.15	<0.001	0.61	0.23
A+B:P	3.33	3.39	3.58	3.54	0.16	<0.001	0.86	0.23
Ammonia-N, mg/dL	4.38	5.04	4.72	4.93	0.61	0.70	0.16	0.47

Ruminal pH and Fermentation

No significant interactions ($P > 0.10$) existed between uNDF240om and peNDF for any measure of ruminal pH (Table 9). Overall, uNDF240om had a much larger effect on ruminal pH than particle size. Cows fed the lower uNDF240om diets had lower mean pH ($P = 0.006$), lower minimum pH ($P = 0.03$), and more time with pH <5.5 ($P = 0.03$). Additionally, the variation in rumen pH (SD of pH; $P = 0.08$) and area under the curve for pH <5.8 tended ($P = 0.06$) to be greater for cows fed lower rather than higher uNDF240om diets. The lower and more variable ruminal pH conditions for cows fed less uNDF240om would be expected given the larger pdNDF fraction and therefore more fermentable substrate (especially at earlier fermentation time points; Table 4) for ruminal fibrolytic bacteria (Van Soest, 1994). A factor potentially contributing to the relatively small magnitude of difference in ruminal pH between lower and higher dietary uNDF240om was the larger inclusion of beet pulp in the lower uNDF240om diets. McBurney et al. (1983) determined that beet pulp had a greater cation exchange capacity (i.e., buffering capacity) because of its elevated pectin content, which serves to mitigate decline in ruminal pH.

Unlike uNDF240om, dietary peNDF had little effect on measures of ruminal pH, other than increasing ($P = 0.008$) the maximum pH. Response of various ruminal pH measures to dietary peNDF has been reported to be variable because it is related not only to salivary buffering elicited by longer particles, but also to VFA production and absorption from the rumen (Beauchemin, 2018). Factors beyond particle size, such as carbohydrate fermentability, passage rates, and health of ruminal epithelium, affect VFA concentrations. Looking to the future, recent research suggests that monitoring ruminal pH variability may be more valuable for dairy management systems than any individual pH threshold value (Heirbaut et al., 2022). In this regard, higher uNDF240om diets tended ($P = 0.08$) to reduce ruminal pH variation because they limited fermentable aNDFom, although they simultaneously constrained DMI if particle size was too large.

There were no significant interactions between uNDF240om and peNDF for VFA concentration, molar percentages of VFA, or ammonia-N ($P > 0.10$; Table 10). Compared with higher uNDF240om, lower dietary uNDF240om significantly increased ($P = 0.02$) total VFA concentration (related to lower pH as discussed), tended ($P = 0.08$) to reduce acetate percentage, increased ($P < 0.001$) propionate percentage, and lowered ($P < 0.001$) the acetate-to-propionate and acetate + butyrate-to-propionate ratios. These changes in VFA concentration and percentages tracked with our observed changes in milk fat and the greater fermentability of diets containing less uNDF240om and more pdNDF. Previous research has found higher total VFA concentration and lower A:P ratios to be associated with the greater fermentability of finer chopped, more degradable diets (Teimouri Yansari et al., 2004). These differences in A:P ratios aid in explaining our observed milk fat percentage differences. The increase in milk fat percentage with the high uNDF240om diets was associated with greater A:P ratios as previously discussed by Miller et al. (2021) for similar types of diets. Our VFA results also agree with Vieira et al. (2025), who fed corn silage–based diets and observed a linear rise in A:P ratio as dietary uNDF240om increased.

Branched-chain VFA, isobutyrate and isovalerate, increased for cows fed the higher versus lower uNDF240om diets. Likewise, higher dietary peNDF tended to increase ruminal valerate ($P = 0.06$) and isovalerate ($P = 0.08$) molar percentages. Lower molar percentages of branched-chain VFA for cows fed less uNDF240om (and higher, more fermentable pdNDF) likely reflect greater aNDFom degradation and the fact that ruminal fibrolytic microbes require these VFA for normal growth (Gehman et al., 2008).

Table 11. Ruminal digesta characteristics, pool size, and turnover for lactating Holstein cows (n = 8) fed diets differing in uNDF240 and peNDF

	Low uNDF240		High uNDF240			*P*-value		
Item	Low peNDF	High peNDF	Low peNDF	High peNDF	SE	uNDF	peNDF	uNDF × peNDF
Ruminal digesta volume, L	110	111	116	111	4.2	0.31	0.52	0.25
Ruminal digesta mass, kg	95	95	100	97	3.7	0.16	0.45	0.38
Ruminal density, kg/L	0.86	0.86	0.87	0.87	0.01	0.35	0.83	0.55
Ruminal pool, kg								
OM	12.67	12.34	12.94	12.39	0.53	0.59	0.14	0.71
aNDFom[1]	8.18	7.93	8.72	8.39	0.36	0.02	0.15	0.86
uNDF240om[2]	3.83	3.72	4.51	4.42	0.17	<0.001	0.34	0.91
Starch	0.40	0.37	0.36	0.31	0.03	0.07	0.19	0.67
Ruminal turnover rate, %/h								
OM	8.7	8.8	8.4	8.0	0.4	0.05	0.45	0.37
aNDFom	4.4	4.4	4.2	3.9	0.2	0.01	0.30	0.31
uNDF240om	2.8	2.8	3.0	2.7	0.1	0.32	0.27	0.21
Starch[3]	123.1 (91.3–166.1)	154.6 (114.6–208.5)	121.0 (89.7–163.2)	154.3 (114.4–208.0)	—	0.94	0.09	0.96
Ruminal turnover time, h								
OM	11.8	11.8	12.1	12.9	0.6	0.06	0.28	0.33
aNDFom	23.2	23.2	24.4	26.1	1.3	0.02	0.24	0.28
uNDF240om	37.4	37.1	34.6	37.0	1.8	0.23	0.37	0.28
Starch	1.4	1.3	1.3	1.3	0.1	0.41	0.48	0.93

[1]Amylase- and sodium sulfite-treated NDF, ash-corrected.

[2]Undegraded NDF after 240 h of in vitro fermentation, ash-corrected.

[3]95% CI in parentheses.

Ruminal Digesta Characteristics and Apparent Total-Tract Nutrient Digestibility

There was no interaction between uNDF240om and peNDF for any measure of ruminal digesta characteristics or turnover ($P > 0.10$; Table 11). Higher dietary uNDF240om, however, increased the ruminal pool size of aNDFom by 0.6 kg and uNDF240om by 0.8 kg versus lower uNDF240om diets. In contrast, dietary peNDF did not affect ruminal pool size of any fiber fraction ($P > 0.10$). The ruminal aNDFom pool size of 8.6 kg observed when cows were fed 11.5% dietary uNDF240om is close to what is thought to be a maximum ruminal fill for NDF (Cotanch et al., 2014; Miller et al., 2021). Responses in ruminal aNDFom pools help to explain our DMI responses because smaller particle size with higher uNDF240om appears to have partially released the fill constraint, allowing greater DM and aNDFom intake.

As with pool size, dietary peNDF had no effect on ruminal turnover rate or time ($P > 0.10$; Table 11). However, greater uNDF240om significantly reduced ($P = 0.01$) ruminal turnover rate of aNDFom thereby increasing turnover time by about 2 h ($P = 0.02$) versus the lower uNDF240om diet. Similarly, Miller et al. (2021) fed corn silage–based diets to lactating cows and observed that lower forage NDF degradability and higher dietary uNDF240om content increased ruminal pool size of aNDFom and slowed its turnover. Overall, in our study and that of Miller et al. (2021), dietary uNDF240om content of 10% or greater reduced DMI and was associated with an aNDFom pool size of approximately 8.6 kg that turned over more slowly.

Ruminal turnover rate and turnover time of starch was not different among the 4 treatment diets. In general, the values are similar, or higher, in magnitude to previous research with corn silage–based diets (92.2 to 103.1, Ivan et al., 2005; 69.6 to 99.4, Farmer et al., 2014) and reflect the high fermentability of the starch sources within our diets such as high moisture corn (from silage), steam-flaked corn grain, and finely ground corn meal.

Apparent Total-Tract Nutrient Digestibility

There was no interaction between uNDF240om and peNDF for apparent total-tract digestibility of OM ($P = 0.73$), aNDFom ($P = 0.51$), pdNDF ($P = 0.50$), or starch ($P = 0.89$). In addition, the main effects themselves had little effect on apparent total-tract nutrient digestibility (Table 12). Dietary uNDF240om did significantly ($P = 0.003$) affect apparent total-tract digestibility of pdNDF, with the 11.5% uNDF240om diets increasing digestibility of pdNDF by 2.7 percentage units versus the 8.8% uNDF240om diets. Greater digestibility of the pdNDF fraction for cows fed higher versus lower uNDF240om can be explained by the greater ruminal aNDFom retention time. Previous research has indicated a positive relationship between extended ruminal retention and greater pdNDF digestibility (Harper and McNeill, 2015). Additionally, all

Table 12. Apparent total-tract nutrient digestibility (%) for lactating Holstein cows (n = 14) fed diets differing in uNDF240 and peNDF[1]

	Low uNDF240		High uNDF240			*P*-value		
Item	Low peNDF	High peNDF	Low peNDF	High peNDF	SE	uNDF	peNDF	uNDF × peNDF
OM	70.2	68.4	71.0	69.8	0.9	0.25	0.11	0.73
aNDFom[2]	51.2	51.1	52.0	50.8	0.8	0.74	0.41	0.51
Potentially degradable NDF[3]	70.8	71.7	74.0	73.7	0.8	0.0033	0.72	0.50
Starch	98.3	98.5	98.6	98.8	0.2	0.03	0.13	0.89

[1]Values are ash-corrected.

[2]Amylase- and sodium sulfite-treated NDF, ash-corrected.

[3]aNDFom − uNDF240om (Waldo et al., 1972).

diets contained in the concentrate mix a yeast culture that may have numerically increased ruminal NDF degradability based on some previous research (White et al., 2008).

CONCLUSIONS

We observed an interaction between uNDF240om and peNDF for DMI, aNDFom intake, and uNDF240om intake. Although particle size had no effect on intake at 8.8% dietary uNDF240om, when uNDF240om was 11.5% of DM, larger particle size decreased DMI by 2.5 kg/d. Greater dietary uNDF240om and peNDF decreased milk yield, and increased milk fat. When combining pef and uNDF240om into a single metric (i.e., peuNDF240), we observed a pattern of response across diets whereby DMI, ECM, chewing time, ruminal pH, and VFA tracked with peuNDF240 regardless of individual changes in uNDF240om or peNDF. Future research needs to assess the usefulness of this measure, or some similar measure, that combines NDF, NDF (un)degradability, and particle size. When dietary uNDF240om is high enough to elicit gut fill constraints for corn silage-based diets (e.g., 11.5% of DM), coarse chopping reduces DMI whereas finer chopping allows DMI similar to more moderate dietary uNDF240om (e.g., 8.8% of DM).

NOTES

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Nonstandard abbreviations used: A = acetate; aNDFom = NDF measured using α-amylase and expressed on an OM basis; AUC = area under the curve; B = butyrate; CNCPS = Cornell Net Carbohydrate and Protein System; FA = fatty acids; HP = higher peNDF; HU = higher uNDF240; HU-HP = higher uNDF240om and higher peNDF concentrations; HU-LP = higher uNDF240om and lower peNDF concentrations; LP = lower peNDF; LU = lower uNDF240; LU-HP = lower uNDF240om and higher peNDF concentrations; LU-LP = lower uNDF240om and lower peNDF concentrations; P = propionate; pdNDF = potentially degradable NDF; pef = physical effectiveness factor; peNDF = physically effective NDF; peuNDF240 = physically effective uNDF240; UNDF = undegraded NDF; uNDF240 = undegraded NDF at 240 h of in vitro fermentation; uNDF240om = uNDF240 measured on an OM basis.

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ORCIDS

W. A. Smith, https://orcid.org/0009-0008-1801-241X
K. Ishida, https://orcid.org/0000-0002-1706-7098
H. M. Dann, https://orcid.org/0000-0002-7372-7011
C. S. Ballard, https://orcid.org/0000-0002-1371-2256
M. D. Miller, https://orcid.org/0000-0002-1448-6289
S. Y. Morrison, https://orcid.org/0000-0001-5209-2283
R. J. Grant https://orcid.org/0000-0001-5797-9916